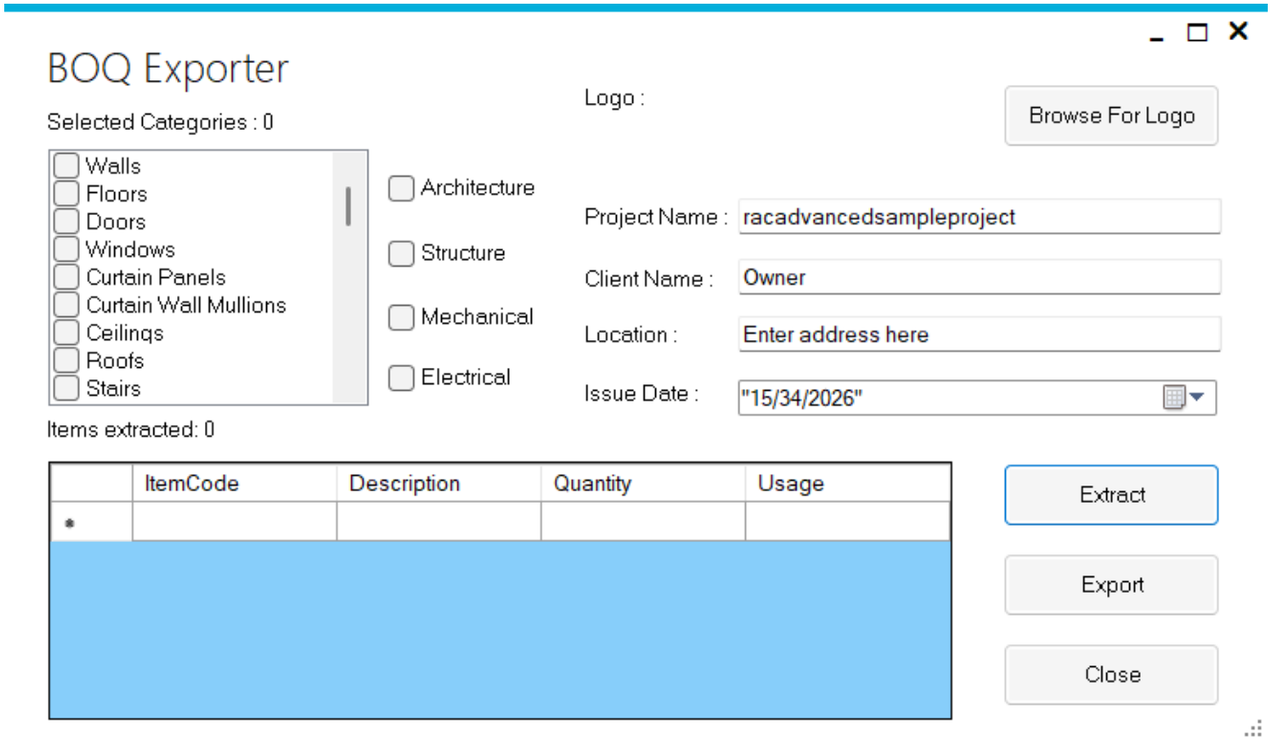
Project: BOQ Exporter

KAITECH

ALLAMARCHCODEDEV

Round: R09

Developer: Mahmoud Amr Allam



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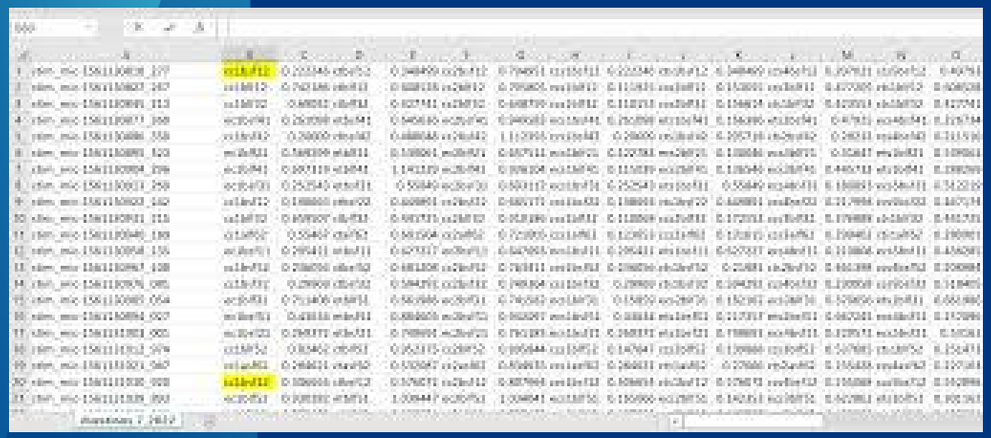
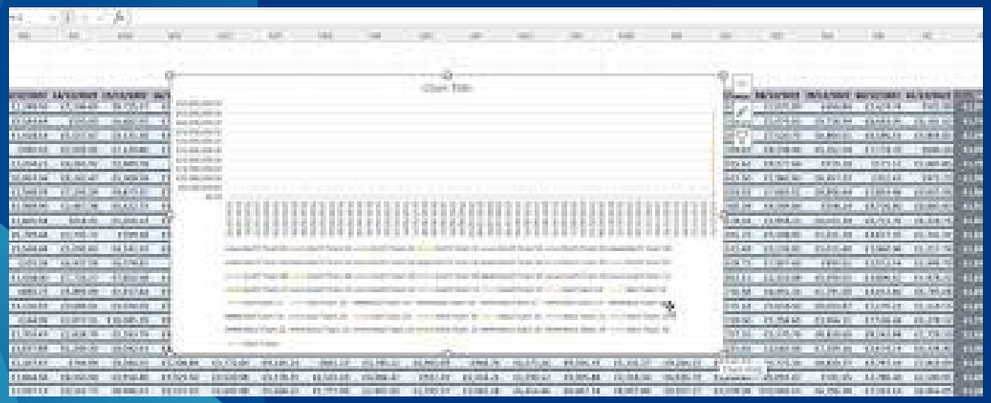
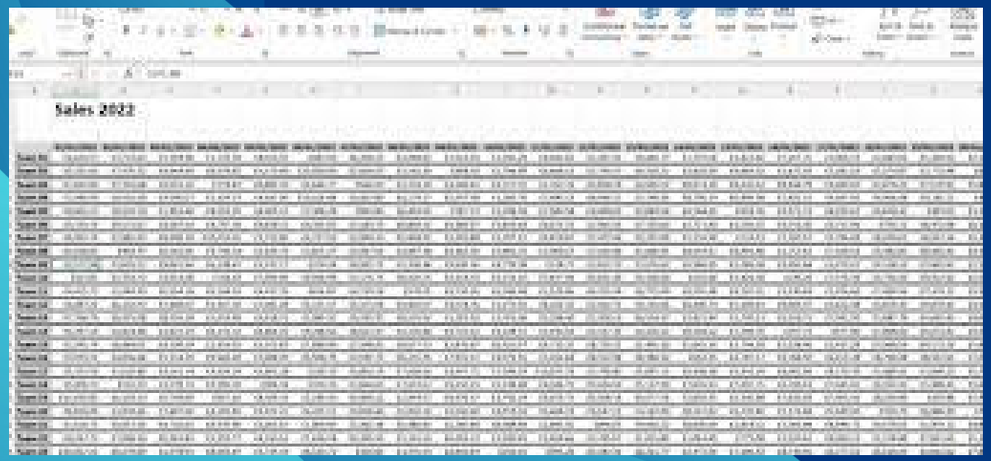
Problem Statement







QUANTITY TAKEOFF IN REVIT IS MANUAL AND ERROR-PRONE.

LACK OF STANDARDIZED EXCEL EXPORTS.

NO VISUAL SUMMARIES OR TRACEABILITY.

Solution Overview

Revit

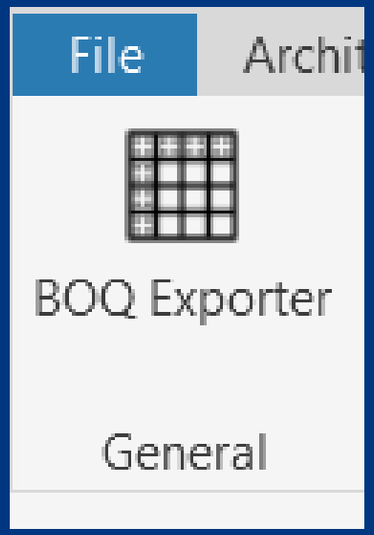
BOQ Exporter

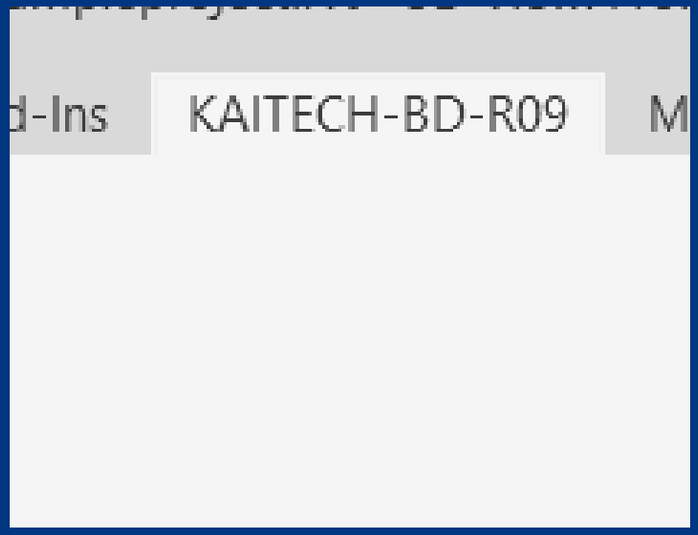
Excel Sheet

BOQ Exporter automates extraction directly from Revit models

Provides structured Excel output with metadata and charts.

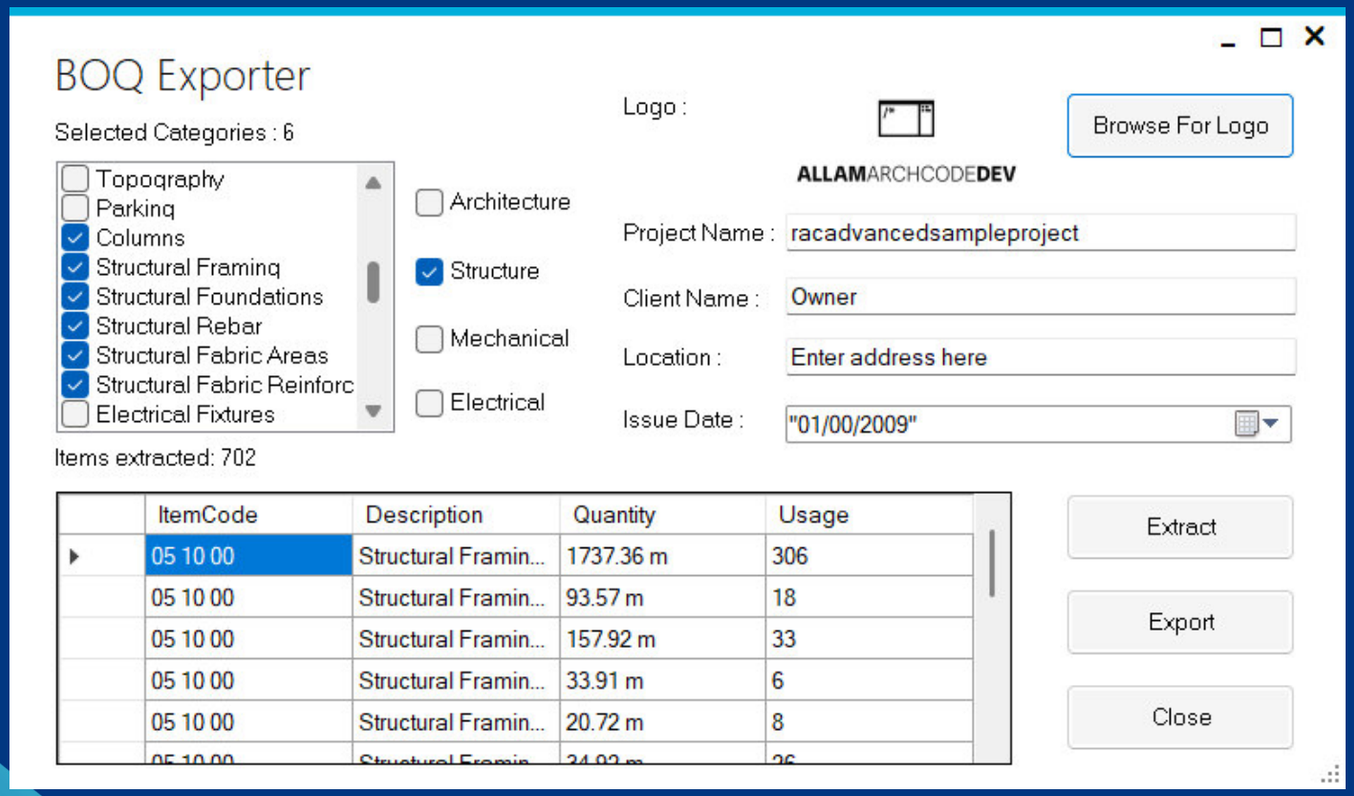
Saves time and reduces errors





UI Features

- Category checklist grouped by discipline
- Metadata entry (Project, Client, Location, Date)
- Logo upload and preview
- Preview before export
- UI state persistence



Code Structure

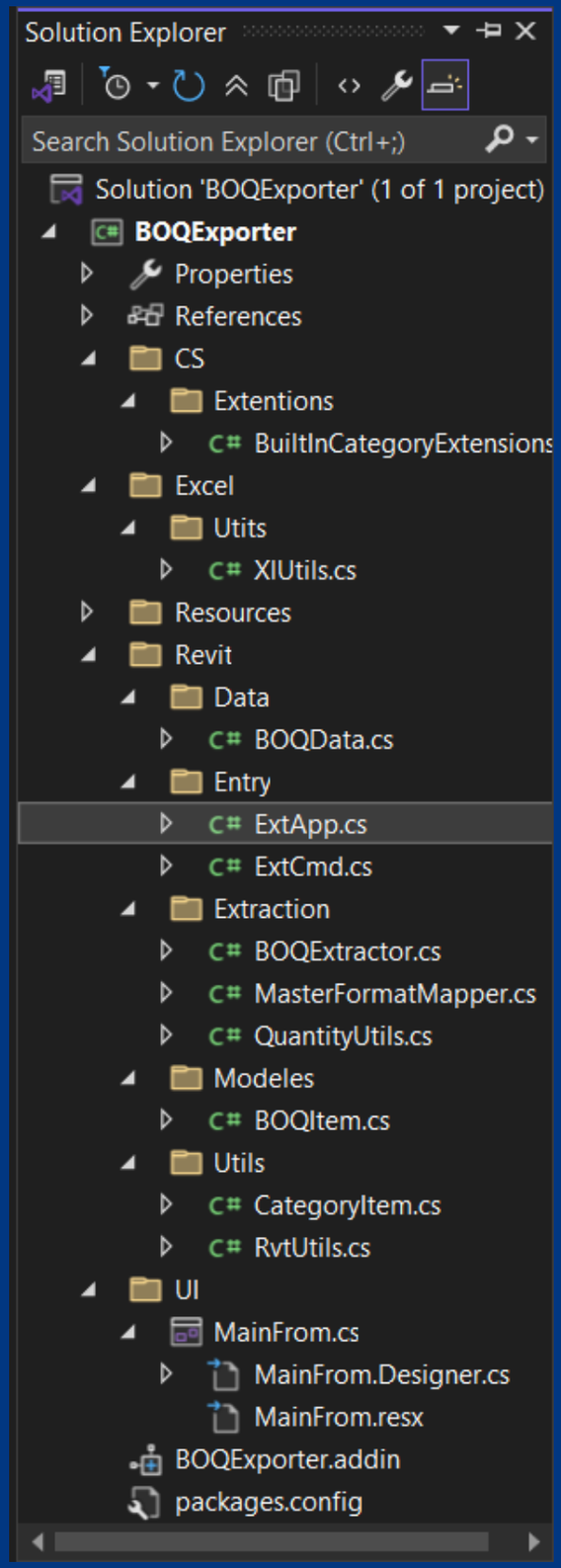
Revit.Entry: ExtApp, ExtCmd (startup and command execution)

Revit.Extraction: BOQExtractor, MasterFormatMapper, QuantityUtils

Revit.Data: BOQData, BOQItem (data models)

UI: MainFrom (Windows Forms interface)

Excel.Utils: XlUtils (Excel formatting and styling)



Extraction Logic

- Uses FilteredElementCollector across 40+ categories
- Applies MasterFormat mapping for item codes
- Converts units to metric (m, m^2, m^3)
- Displays extracted items in UI before export

ItemCode	Description	Quantity	Usage
04 20 00	Walls: Exterior - ...	783.61 m^2	8
08 44 00	Walls: Exterior C...	2446.65 m^2	18
08 44 00	Walls: Pavillion ...	129.99 m^2	3
09 00 00	Walls: Interior - 1...	2887.57 m^2	122
09 00 00	Walls: Generic - ...	379.64 m^2	4
09 00 00	Walls: Storefront	176.15 m^2	12

BOQExtractor.cs

BOQExporter | BOQExporter.Revit.Extraction.BOQExtractor

```
{
    /// <summary>
    /// extract BOQ items from the Revit document over 58 cat.
    /// </summary>
    /// <param name="doc"></param>
    1 reference
    public static void Extract(Document doc , HashSet<BuiltInCategory> allowedCategories)
    {
        BOQData.Initialize();

        // to check if the category is allowed
        bool IsAllowed(BuiltInCategory bic) => allowedCategories.Contains(bic);


        //------Architectural Elements Extraction------20

        #region Walls Extraction
        if (IsAllowed(BuiltInCategory.OST_Walls))
        {
            var walls = new FilteredElementCollector(doc)
                .OfCategory(BuiltInCategory.OST_Walls)
                .WhereElementIsNotElementType()
                .Cast<Wall>();

            foreach (var wall in walls)
            {
                double area = wall.get_Parameter(BuiltInParameter.HOST_AREA_COMPUTED)?.AsDouble() ?? 0;
                double areaM2 = QuantityUtils.ToSquareMeters(area);

                string code = MasterFormatMapper.GetCode(wall);
                string desc = wall.Name;
                string cate = wall.Category?.Name ?? "_";
                BOQData.Items.Add(new BOQItem(cate, code, desc, Math.Round(areaM2, 2), "m²"));
            }
        }
        #endregion
```

MasterFormatMapper.cs

BOQExporter | BOQExporter.Revit.Extraction.MasterFormatMapper

```
using Autodesk.Revit.DB;
using System.Collections.Generic;

namespace BOQExporter.Revit.Extraction
{
    /// <summary>
    /// this static class provides functionality to map Revit element categories and type names
    /// </summary>
    50 references
    internal static class MasterFormatMapper
    {
        #region MasterFormat Supported Categories
        // Dictionary: BuiltInCategory → (keyword → MasterFormat code)
        private static readonly Dictionary<BuiltInCategory, Dictionary<string, string>> MappingRules =
            new Dictionary<BuiltInCategory, Dictionary<string, string>>
            {
                // ---------------- ARCHITECTURE ----------------
                { BuiltInCategory.OST_Walls, new Dictionary<string, string>
                    {
                        { "structural", "03 30 00" }, // Cast-in-Place Concrete (structural walls)
                        { "concrete", "03 30 00" },
                        { "masonry", "04 20 00" },
                        { "brick", "04 21 00" },
                        { "curtain", "08 44 00" },
                        { "gypsum", "09 29 00" },
                        { "drywall", "09 29 00" },
                        { "metal stud", "09 22 00" },
                        { "cmu", "04 22 00" },
                        { default", "09 00 00" }
                    }
                },

                { BuiltInCategory.OST_Floors, new Dictionary<string, string>...
                },
```

Excel Output

- **BOQ Report**: itemized table with codes, descriptions, quantities, totals
- **BOQ Summary Report**: usage counts for traceability
- **Category Analysis**: distribution table + pie chart

Project Name: racadvancedsampleproject
Client: Owner
Location: Enter address here
Date: 01-04-2009

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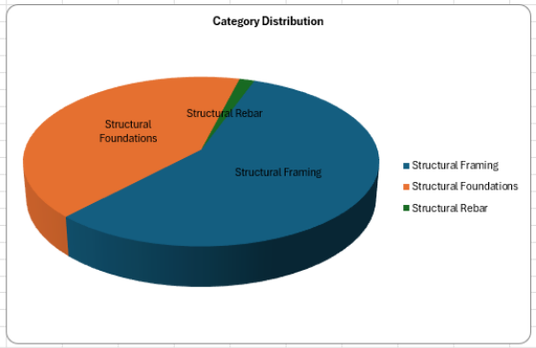
BOQ Report

Item No.	Item Code	Description	Quantity	Unit	Unit Rate	Total Cost	Remarks
1	05 10 00	Structural Framing: 400 x 800mm	1737.36	m	0.00 EGP	0.00 EGP	
2	05 10 00	Structural Framing: 300 x 600mm	93.57	m	0.00 EGP	0.00 EGP	
3	05 10 00	Structural Framing: 600 x 900	157.92	m	0.00 EGP	0.00 EGP	
4	05 10 00	Structural Framing: UB305x165x40	33.91	m	0.00 EGP	0.00 EGP	
5	05 10 00	Structural Framing: UB254x102x28	20.72	m	0.00 EGP	0.00 EGP	
6	05 10 00	Structural Framing: HSS114.3X8.6	34.92	m	0.00 EGP	0.00 EGP	
7	03 10 00	Structural Foundations: 800 x 1800 x 900mm	48	ea	0.00 EGP	0.00 EGP	
8	03 10 00	Structural Foundations: 400mm Diameter	96	ea	0.00 EGP	0.00 EGP	
9	03 10 00	Structural Foundations: 2000 x 2000 x 900mm	30	ea	0.00 EGP	0.00 EGP	
10	03 10 00	Structural Foundations: 500mm Diameter	120	ea	0.00 EGP	0.00 EGP	
11	03 20 00	Structural Rebar: 16M : Shape M_T1	99.15	m	0.00 EGP	0.00 EGP	
12	03 20 00	Structural Rebar: 16M : Shape M_00	50.1	m	0.00 EGP	0.00 EGP	
13	03 20 00	Structural Rebar: 13M : Shape M_00	66.35	m	0.00 EGP	0.00 EGP	
14	03 21 00	Structural Rebar: 13M : Shape Rebar Shape 1	1.38	m	0.00 EGP	0.00 EGP	
15	03 20 00	Structural Rebar: 19M : Shape M_00	6215.4	m	0.00 EGP	0.00 EGP	
					Subtotal:	0.00 EGP	

Category Analysis

Label	Value	Percentage
Structural Framing	397	57%
Structural Foundations	294	42%
Structural Rebar	11	2%



BOQ Summary Report

Item Code	Category	Description	Quantity	Unit	Usage Count
05 10 00	Structural Framing	400 x 800mm	1737.36	m	306
05 10 00	Structural Framing	300 x 600mm	93.57	m	18
05 10 00	Structural Framing	600 x 900	157.92	m	33
05 10 00	Structural Framing	UB305x165x40	33.91	m	6
05 10 00	Structural Framing	UB254x102x28	20.72	m	8
05 10 00	Structural Framing	HSS114.3X8.6	34.92	m	26
03 10 00	Structural Foundations	800 x 1800 x 900mm	48	ea	48
03 10 00	Structural Foundations	400mm Diameter	96	ea	96
03 10 00	Structural Foundations	2000 x 2000 x 900mm	30	ea	30
03 10 00	Structural Foundations	500mm Diameter	120	ea	120
03 20 00	Structural Rebar	16M : Shape M_T1	99.15	m	1
03 20 00	Structural Rebar	16M : Shape M_00	50.1	m	6
03 20 00	Structural Rebar	13M : Shape M_00	66.35	m	1
03 21 00	Structural Rebar	13M : Shape Rebar Shape 1	1.38	m	2
03 20 00	Structural Rebar	19M : Shape M_00	6215.4	m	1
				Total Items:	702

BOQ Data

Project Name: racadvancedsampleproject
Client: Owner
Location: Enter address here
Date: 01-04-2009

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BOQ Report

Item No.	Item Code	Description	Quantity	Unit	Unit Rate	Total Cost	Remarks
1	05 10 00	Structural Framing: 400 x 800mm	1737.36	m	0.00 EGP	0.00 EGP	
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4	05 10 00	Structural Framing: UB305x165x40	33.91	m	0.00 EGP	0.00 EGP	
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7	03 10 00	Structural Foundations: 800 x 1800 x 900mm	48	ea	0.00 EGP	0.00 EGP	
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9	03 10 00	Structural Foundations: 2000 x 2000 x 900mm	30	ea	0.00 EGP	0.00 EGP	
10	03 10 00	Structural Foundations: 500mm Diameter	120	ea	0.00 EGP	0.00 EGP	
11	03 20 00	Structural Rebar: 16M : Shape M_T1	99.15	m	0.00 EGP	0.00 EGP	
12	03 20 00	Structural Rebar: 16M : Shape M_00	50.1	m	0.00 EGP	0.00 EGP	
13	03 20 00	Structural Rebar: 13M : Shape M_00	66.35	m	0.00 EGP	0.00 EGP	
14	03 21 00	Structural Rebar: 13M : Shape Rebar Shape 1	1.38	m	0.00 EGP	0.00 EGP	
15	03 20 00	Structural Rebar: 19M : Shape M_00	6215.4	m	0.00 EGP	0.00 EGP	
					Subtotal:	0.00 EGP	

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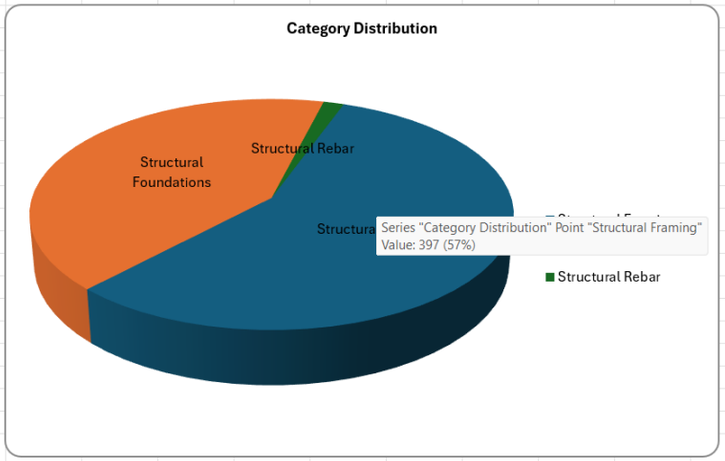

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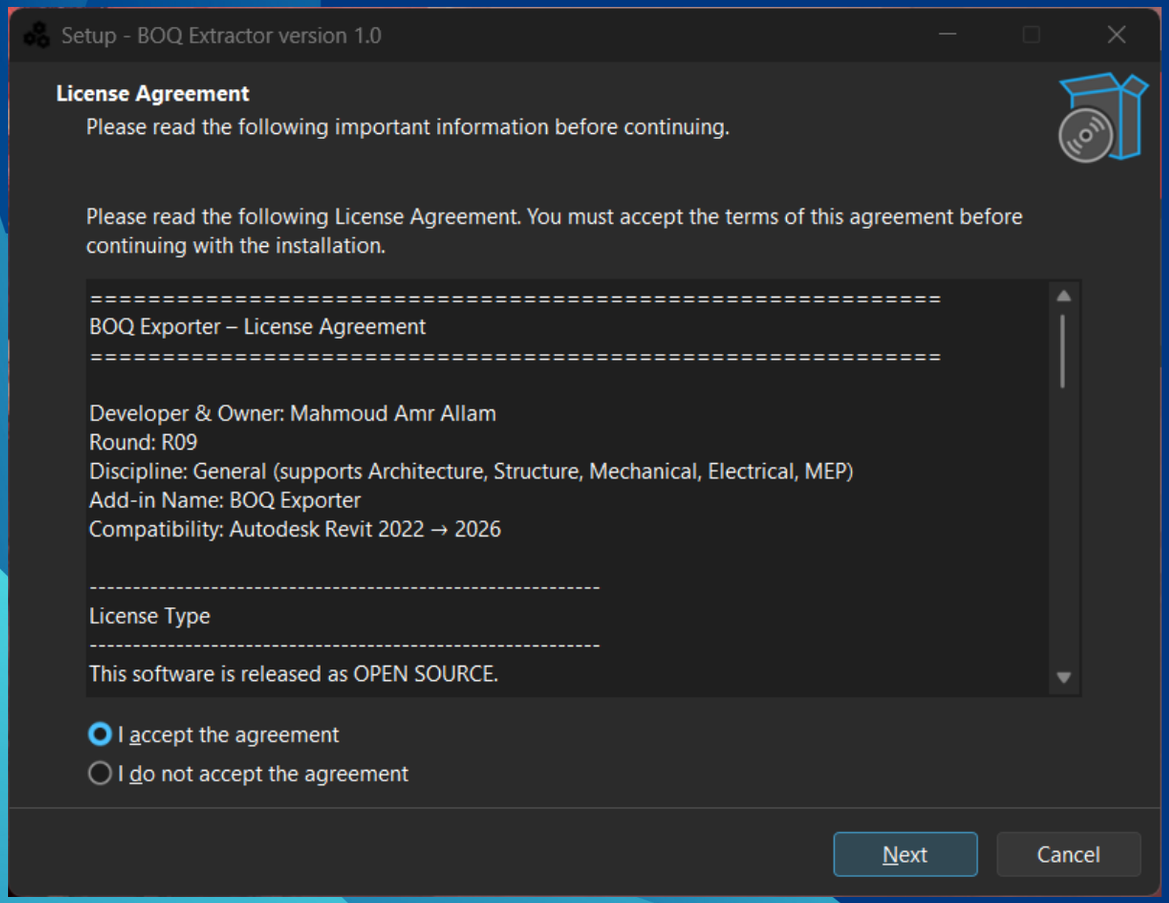
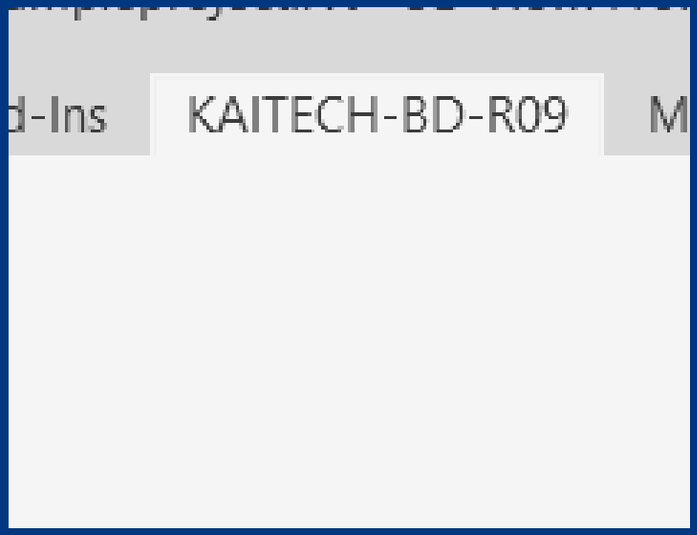
03 10 00	Structural Foundations	800 x 1800 x 900mm	48	ea	48
03 10 00	Structural Foundations	400mm Diameter	96	ea	96
03 10 00	Structural Foundations	2000 x 2000 x 900mm	30	ea	30
03 10 00	Structural Foundations	500mm Diameter	120	ea	120
03 20 00	Structural Rebar	16M : Shape M_T1	99.15	m	1
03 20 00	Structural Rebar	16M : Shape M_00	50.1	m	6
03 20 00	Structural Rebar	13M : Shape M_00	66.35	m	1
03 21 00	Structural Rebar	13M : Shape Rebar Shape 1	1.38	m	2
03 20 00	Structural Rebar	19M : Shape M_00	6215.4	m	1
				Total Items:	702

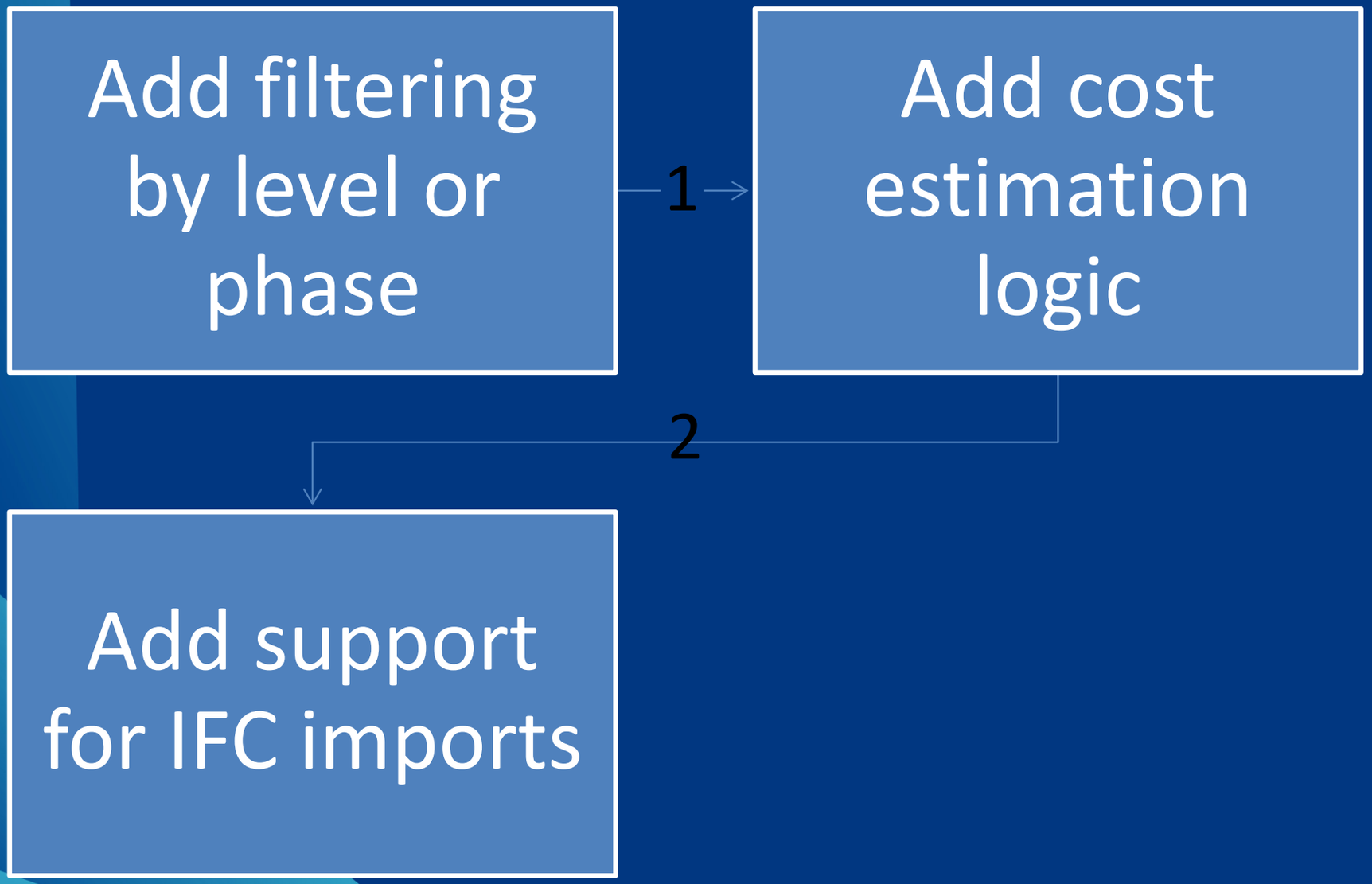
Setup & Compatibility

- One installer supports Revit 2022–2026
- .addin manifest auto-registered
- Tab: KAITECH-BD-R09 → Panel: General





Future Enhancements



Thank You

Contact: Mahmoud Amr Allam

Submission: KT-BD-R09-GP-Mahmoud Amr Allam-BOQExporter